Answer **any one** question in this Section.

**8.** **(a)** **(i)** State two assumptions of the kinetic theory of gases.

**1.** .....

.....

**2.** .....

..... [2]

**(ii)** A sealed container holds a mixture of nitrogen-14 and helium-4 gas molecules at at temperature of 290 K.

**1.** Determine the average kinetic energy of a helium molecule.

average kinetic energy = ..... J [2]

**2.** Explain whether it is possible for the mean square speed of the nitrogen molecules to be different from the helium molecules at the same temperature.

.....

..... [1]

**(b)** **(i)** State what is meant by the internal energy of a system.

.....

.....

..... [1]

**(ii)** In Fig. 8.1 below, place a tick (√) against those changes where the internal energy of a system is increasing.

| | |
|---|---|
| Water freezing at constant temperature | |
| A stone falling under gravity in a vacuum | |
| Water evaporating at constant temperature | |
| Stretching a wire at constant temperature. | |

**Fig. 8.1**

[2]

**(iii)** A cylinder of gas at constant volume is placed under the sun so that its temperature rises.

Explain the changes to the internal energy of the gas.

.....

.....

.....

..... [2]

**(iv)** The volume occupied by 1.00 mol of water at 100 °C is $1.87 \times 10^{-5}$ $m^3$. When the water is vaporised at an atmospheric pressure of $1.03 \times 10^5$ Pa, the water vapour has a volume of $2.96 \times 10^{-2}$ $m^3$.

The latent heat required to vaporise 1.00 mol of water at 100 °C and $1.03 \times 10^5$ Pa is $4.05 \times 10^4$ J.

Determine, for this change of state,

**1.** the work done on the system,

work done = ..... J [2]

**2.** the increase in internal energy of the system.

increase in internal energy = ..... J [2]

**(c)** **(i)** A student states wrongly that temperature measures the amount of thermal energy in a body.

State and explain two observations that show why the statement is incorrect.

.....

.....

.....

..... [2]

**(ii)** Some crushed ice at 0 °C is placed in a funnel together with an electric heater, as shown in Fig. 8.2 below.

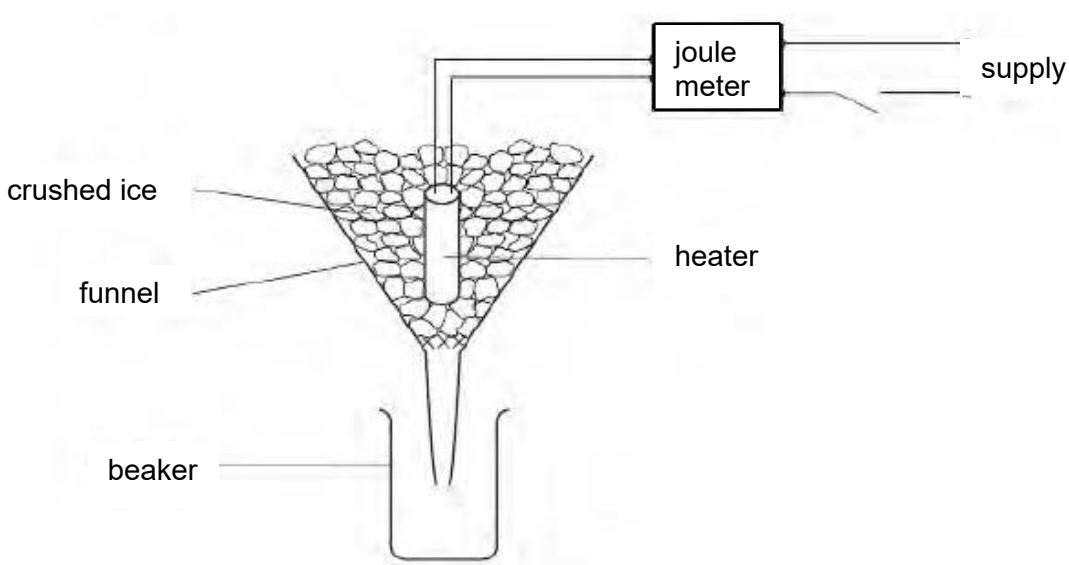


**Fig. 8.2**

The mass of water collected in the beaker in a measured interval of time is determined with the heater switched off. The mass is then found with the heater switched on. The energy supplied to the heater is also measured.

For both measurements of the mass, water is not collected until melting occurs at a constant rate.

The data shown in Fig. 8.3 are obtained.

| | mass of water / g | energy supplied to heater / kJ | time interval / min |
|---|---|---|---|
| heater switched off | 8.3 | 0 | 5.0 |
| heater switched on | 64.7 | 18 | 5.0 |

**Fig. 8.3**

**1.** State why it is necessary to determine the mass of water with the heater switched off.

.....

..... [1]

**2.** Suggest how it can be determined that the ice is melting at a constant rate.

.....

..... [1]

**3.** Calculate a value for the specific latent heat of fusion of ice.

specific latent heat = ..... $kJ\ kg^{-1}$ [2]